Table of Contents

- UW12GVNADefTopic
- UW12GVNA_Debug.5.1
- UW12GVNA_Debug.5.2
- UW12GVNA_about.1.2
- UW12GVNA_about.1.3
- UW12GVNA_about.1.4
- UW12GVNA_cal.3.1
- UW12GVNA_cal.3.2
- UW12GVNA_programming.4.1
- UW12GVNA_programming.4.2
- UW12GVNA_programming.4.3
- UW12GVNA_programming.4.4
- UW12GVNA_programming.4.5
- UW12GVNA_programming.4.6
- UW12GVNA_theory.2.1
- UW12GVNA_theory.2.2
- UW12GVNA_theory.2.3

UW12GVNADefTopic

<	>

About the UltraWave 12G VNA

About the UltraWave 12G VNA

The UltraWave 12G Vector Network Analyzer (VNA) allows you to make s-parameter measurements, which provide a linear model for device behavior.

The UltraWave 12G VNA is used in testing cell phone, wireless network, and other devices operating in the 50MHz to 12GHz frequency range.

This section includes the following topics:

- Features
- Safety Information
- Key Terms

UltraWave 12G VNA information includes the following sections:

- UltraWave 12G VNA Theory Of Operation
- UltraWave 12G VNA Calibration
- UltraWave 12G VNA Programming
- UltraWave 12G VNA Debug Displays

Related Topics

- UltraWave 12G Specifications
- MWVNA (VBT syntax)
- UltraWave 12G DIB Design Guidelines
- About the UltraWave 12G Subsystem
- UltraWave 12G Examples

- Smith Display

<	>

UW12GVNA_Debug.5.1

<	>

UltraWave 12G VNA Debug Displays

UltraWave 12G VNA Debug Displays

The Microwave Debug Display can display one or more instrument blocks for the UltraWave 12G VNA logical instrument. Each instrument block shows the state of an instrument in the context of a site.

This section includes the following topic:

- MW12G VNA Debug Display Controls

The Smith Display also supports debugging s-parameter measurements made with the MWVNA. See Smith Display.

For an overview of all the displays supporting the MW subsystem, see UltraWave 12G Displays.

For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

<	>

UW12GVNA_Debug.5.2

<	>

UltraWave 12G VNA Debug Displays : MW12G VNA Debug Display Controls

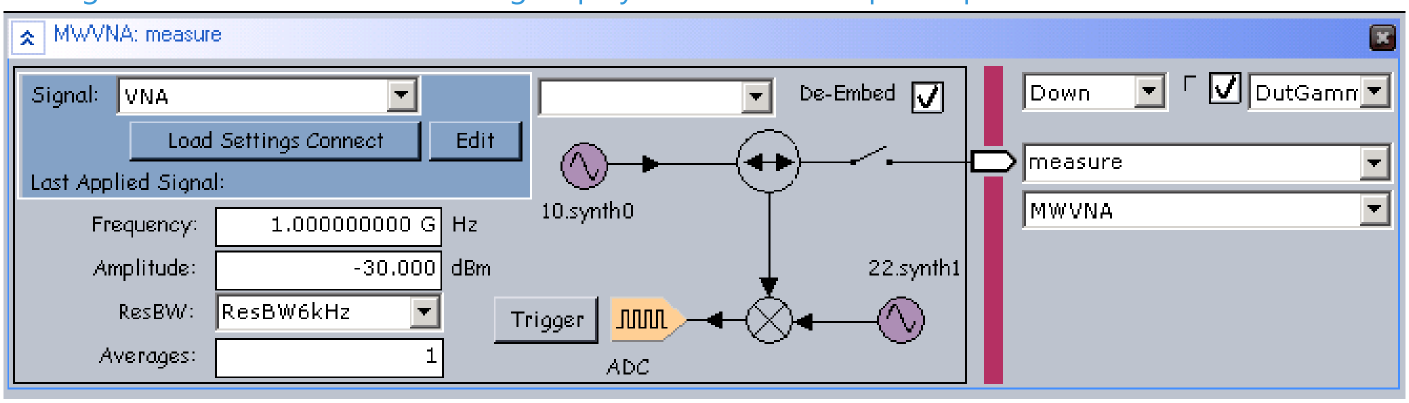
---

MW12G VNA Debug Display Controls

MWVNA instrument display controls allow you to view and set common MWVNA parameters, which are also accessible through Pins VBT language. The MWVNA instrument display shows one-port measurement mode.

For general information about the debug display controls, see UltraWave 12G Displays.

The figure shows the MWVNA debug display block for a one-port s-parameter measurement.



MWVNA Logical Instrument Display

The following controls are available on the MWVNA logical instrument block:

Signal	Displays the selected capture Signal on a drop-down list of available signals.
Load Settings Connect	Loads the selected signal and connects the MWVNA.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.

Last Applied Signal	Name of the last Signal applied to the MWVNA.
De-Embed	Enables or disabled de-embedding. If checked, then the correction specified in MWDIB used, per the pin selected in the display.
Gamma	Enables or disables Gamma. If checked, then the correction specified in MWDIB used, per the pin selected in the display.
Frequency	Displays and sets the frequency of the sine wave sourced to the DUT by the VNA instrument during a s-parameter measurement.
Amplitude	Displays and sets the measurement signal amplitude in dBm.
ResBW	Displays and sets the resolution bandwidth during captures for a measurement.
Averages	Displays and sets the number of captures averaged for an s-parameter measurement.
Trigger	Starts the digitizer capture at the frequency specified in Frequency and the subsequent processing that results in a measured value.
connection relay	The relay control connects or disconnects the VNA logical instrument to the specified pin or channel. Click the relay to open or close the connection.
Bidirectional port	Represents the MW port connected for a measurement, with a pin or channel name for the port in the text box.
synth label	Displays which synthesizer supplies the source signal for a one-port measurement.

☑ **Note:**

If you select de-embedding, the de-embed information must encompass the requested measurement frequency. Otherwise, an error will occur. For example, the de-embed information runs from 1 - 2GHz, but you attempt to measure @ 500MHz. In this case, an error is generated if de-embedding is enabled. For more information, see TheHdw.MWDIB.

< >

UW12GVNA_about.1.2

<	>

About the UltraWave 12G VNA : Features

Features

The UltraWave 12G VNA can measure one-port devices. One-port devices (one RF pin, or differential pair of pins) have one s-parameter, Gamma (Γ), the coefficient of reflection, which maps to impedance.

S-parameters can be used to compensate for impedance mismatch in the DIB or DUT and loss or gain in the DIB which would otherwise affect absolute amplitude accuracy.

For UltraWave 12G hardware specifications, see UltraWave 12G Specifications.

<	>

UW12GVNA_about.1.3

<	>

About the UltraWave 12G VNA : Safety Information

Safety Information

When operating the Microwave Source instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

<	>

UW12GVNA_about.1.4

<	>

About the UltraWave 12G VNA : Key Terms

Key Terms

AWG	Arbitrary Waveform Generator
baseband	Low frequency information contained in a modulated signal
I signal	In-phase (real) component of a baseband signal
IF	Intermediate Frequency
LO	Local Oscillator
Q signal	Quadrature (imaginary) component of a baseband signal
RF	Radio Frequency

For more MW terms and definitions, see Key Terms in About the UltraWave 12G Subsystem.

<	>

UW12GVNA_cal.3.1

<	>

UltraWave 12G VNA Calibration

UltraWave 12G VNA Calibration

The UltraWave 12G VNA instrument supports one-port s-parameter measurements. A signal path connects each VNA instrument (board) internal input/output terminal to a DUT port. The signal path affects measurement accuracy. To increase measurement accuracy, the VNA must be calibrated with all or part of the signal path.

☑ **Note:**

VNA calibration has no effect on UltraWave 12G Source or UltraWave 12G Receiver calibration.

For a high-level description of VNA calibration, see Overview of UltraWave 12G VNA Calibration

<	>

UW12GVNA_cal.3.2

<	>

UltraWave 12G VNA Calibration : Overview of UltraWave 12G VNA Calibration

Overview of UltraWave 12G VNA Calibration

The following topics explain basic VNA calibration concepts:

- Calibration Planes
- VNA S-Parameter Calibrations
- Calibrated Measurements and Calibration Data

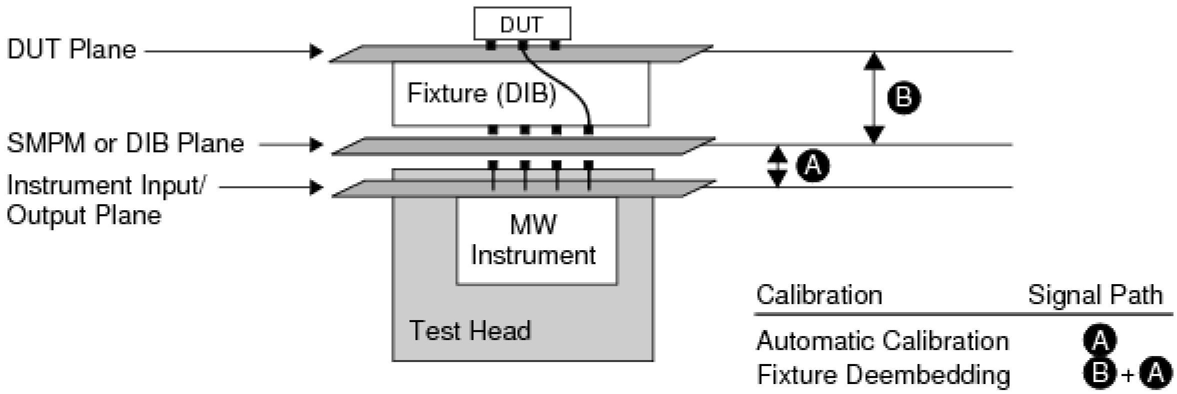
Calibration Planes

The term calibration plane describes where measurement accuracy is defined in a signal path. Ideally, the DUT ports would connect directly to the VNA instrument terminals, with no intervening connection signal path. In this ideal case, the instrument plane and the DUT plane would be the same.

In a practical test system, the three important calibration planes for VNA calibration are:

Instrument input/output plane	Located at the input/output connections of the VNA instrument (board) hardware in the MW subsystem.
SMPM plane	Located at the test head SMPM connector interface for the test fixture or DIB. Also called the DIB plane.
DUT plane	Located at the test fixture socket terminals closest to the DUT.

The figure illustrates the locations of the calibration planes.



MW VNA S-Parameter Calibration

VNA S-Parameter Calibrations

An s-parameter calibration is a way to obtain data to allow calibrated s-parameter measurements. IG-XL supports the following types of VNA s-parameter calibration, listed in order of increasing accuracy:

Automated calibration with Cal DIB fixture	Calibrates the UltraWave 12G VNA from the Maintenance Environment using s-parameter data you obtain from an automated process with the Cal DIB fixture. Calibrates the UltraWave 12G VNA to the SMPM plane.
Fixture deembedding	Calibrates the UltraWave 12G VNA using s-parameter data you obtain from bench measurements (or other characterization) of the fixture (DIB), cascaded with the automated calibration data. Calibrates the UltraWave 12G VNA to the DUT plane.

See the previous figure for an illustration of the signal path for each kind of calibration.

Calibrated Measurements and Calibration Data

For each kind of VNA calibration, the corresponding calibration process creates data for the signal path up to the calibration plane. Once calibration data is available, IG-XL can use it to make calibrated measurements:

Data from automated calibration	During a measurement, the VNA instrument driver gets calibration data from the \IG-XL\Cal\MWMeasure folder. The data is for the latest VNA calibration performed either at the factory or in the field.

Data from fixture deembedding	Deembed data is stored in either a distinct file or an EEPROM. You then use a MWDIB object in your test program workbook and the CurrentLabel property of the MWDIB object to specify the deembed data for a calibrated path. The MWVNA software cascades the automatic calibration data with the fixture deembed data. For more information, see TheHdw.MWDIB.

When using the .SubFixture VBT node in MWDIB, each subfixture level includes an index. The higher the index, the further away from the test port from an embedding standpoint. The MWDIB cascades all subfixtures (in increasing index order away from the test port) to determine the final corrections.

For example, you have the following:

- A scalar loss on your PIB stored in a .s2p file
- A scalar loss of 0.5dB on your probe card for pin RX1 at 2GHz on site 0

The VBT code would look like the following:

```
TheHdw.MWDIB.Pins("RX1").SubFixture(0).Read("./paths/fixture1.s2p", 0, label)
TheHdw.MWDIB.Pins("RX1").SubFixture(1).AddScalar(-0.5, 2000000000#, 0, label)
```

Here, .SubFixture(0) is closest to the calibration plane/tester. The combined .s2p file and scalar value deembed both sections of the path.

For more information, see TheHdw.Pins.SubFixture.

UW12GVNA_programming.4.1

<	>

UltraWave 12G VNA Programming

UltraWave 12G VNA Programming

The UltraWave 12G VNA is a MW logical instrument in the IG-XL environment. In general, you program the UltraWave 12G VNA like the other UltraWave 12G instruments.

The following topics are available:

- Programming Concepts
- Programming Steps
- UltraWave 12G VNA Pattern Control

Before writing a program using the VNA, see IG-XL Programming for UltraWave 12G Instruments.

For more information, see:

- MWVNA (VBT Syntax Reference)
- UltraWave 12G Examples

<	>

UW12GVNA_programming.4.2

<	>

UltraWave 12G VNA Programming : Programming Concepts

Programming Concepts

UltraWave 12G VNA Reset Values

The table lists the local reset values for the UltraWave 12G VNA.

UltraWave 12G VNA Reset Values

Setting	Reset Value
Amplitude	-30 dBm
Averages	1
Frequency	No value
Resolution BW	6kHz
Connections	Disconnected

<	>

UW12GVNA_programming.4.3

<	>

UltraWave 12G VNA Programming : Programming Steps

Programming Steps

The following example describes how to build a test program that uses the Microwave VNA instrument to perform a one-port s-parameter measurement. This example connects the instrument port to one RF DUT pin.

1. Create a pin map:
2. Create a channel map:
3. Type the following code for performing an s-parameter measurement:

```
Public Function MeasureSParameter()
Dim Freq As Double
Dim SParameter As New PinListData
    Freq = 5230000000#
    ' Create, apply, settle, and then trigger a Signal for the measurement
    With TheHdw.MWVNA("RF_Pin").Signals.Add("SParameter")
        .Frequency = Freq
        .LoadSettingsConnect
    TheHdw.Wait 0.01
        .Trigger
    End With
    ' Wait for the measurement to complete and then read back the results
    ' of the s-parameter measurement
```

```
    With TheHdw.MWVNA("RF_Pin")
        .WaitForCaptureDone
        SParameter = .Signals("SParameter").GetGamma
    End With
End Function
```

<	>

UW12GVNA_programming.4.4

<	>

UltraWave 12G VNA Programming : UltraWave 12G VNA Pattern Control

UltraWave 12G VNA Pattern Control

You can control an UltraWave 12G VNA instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- UltraWave 12G VNA Pattern Microcodes
- UltraWave 12G VNA ASCII Pattern Syntax

☑ **Note:**

To use pattern microcodes to control the UltraWave 12G VNA on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

<	>

UW12GVNA_programming.4.5

<	>

UltraWave 12G VNA Programming : UltraWave 12G VNA Pattern Control : UltraWave 12G VNA Pattern Microcodes

UltraWave 12G VNA Pattern Microcodes

A microcode controls a single function, such as starting the capture of a signal. A series of extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is captured. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start capturing a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used for issuing commands to an UltraWave 12G VNA instrument from a pattern.

UltraWave 12G VNA Pattern Microcodes

Microcode	Functional Description
Disconnect	Instructs an UltraWave 12G VNA to reset all connections to their disconnected state.
Trig <SignalName>	The instrument immediately starts capturing the named capture signal. The Trig microcode takes an optional argument, which is the name of the signal to be captured. If the signal name is unspecified, the default signal is used (see TheHdw.MWVNA.Pins.Signals.DefaultSignal).

Enable_Inst_Cond	Enables the instrument condition. The UltraWave 12G VNA asserts the instrument condition to the pattern generator at the completion of a capture.
Disable_Inst_Cond	Disables an UltraWave 12G VNA from asserting the instrument condition.
LoadSettingsConnect <SignalName>	Instructs an UltraWave 12G VNA to load the settings for the named capture signal and connects the hardware to the DUT. SignalName is required.

Note the following when using microcodes with an MWVNA:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:
  - LoadSettingsConnect: 200 - 400s (typical)
  - Disconnect: <200s (typical)
  - Trig: <10s
- An MWVNA does not use PSets. It provides the equivalent capability using Signals programming in VBT.
- When using an UltraWave logical instrument (such as an MWVNA), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.
  - IG-XL returns an error if you do not follow this guideline.
  - Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see Using UltraWave 12G Instruments with POP.

<	>

UW12GVNA_programming.4.6

<	>

UltraWave 12G VNA Programming : UltraWave 12G VNA Pattern Control : UltraWave 12G VNA ASCII Pattern Syntax

UltraWave 12G VNA ASCII Pattern Syntax

UltraWave 12G VNA pattern microcodes are instructions you can place on an individual vector of a pattern. For that vector, a microcode instructs an UltraWave 12G VNA to perform a particular action, such as triggering a new capture.

You can only use UltraWave 12G VNA microcodes with pins that the instruments statement has specified as MWVNA. For example:

```
instruments = {
    DUT_AOUT:MWVNA;
}
```

For more information, see Instruments Statement in the Pattern Language section.

If a vector contains any UltraWave 12G VNA related microcode other than nop, that vector is placed in SRM.

The format for specifying an UltraWave 12G VNA microcode on a vector is:

(pin-list : MWVNA = microcode)

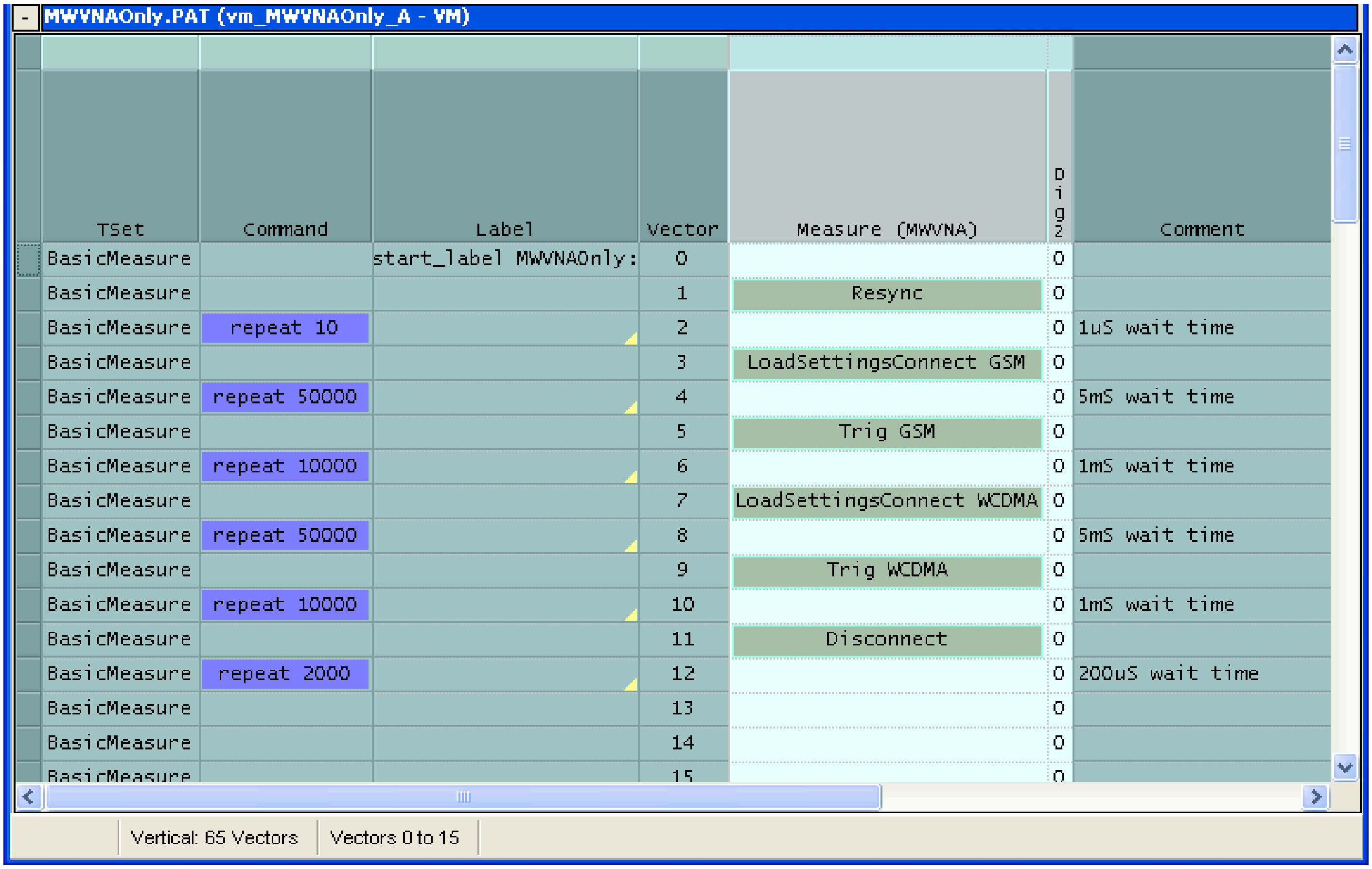
The following code shows an example of ASCII pattern microcode syntax for an MWVNA. The figure shows this pattern in PatternTool.

```
instruments = {
    Measure:MWVNA 1;
}

vm_vector
vm_MWVNAOnly_A ( $tset      , Dig2)
{
start_label MWVNAOnly:
```

```
            > BasicMeasure  0    ;
(Measure:MWVNA = Resync)
            > BasicMeasure  0    ;
repeat 10       > BasicMeasure  0    ; //1uS wait time
(Measure:MWVNA = LoadSettingsConnect GSM)
            > BasicMeasure  0    ;
repeat 50000    > BasicMeasure  0    ; //5mS wait time
(Measure:MWVNA = Trig GSM)
            > BasicMeasure  0    ;
repeat 10000    > BasicMeasure  0    ; //1mS wait time
(Measure:MWVNA = LoadSettingsConnect WCDMA)
            > BasicMeasure  0    ;
repeat 50000    > BasicMeasure  0    ; //5mS wait time
(Measure:MWVNA = Trig WCDMA)
            > BasicMeasure  0    ;
repeat 10000    > BasicMeasure  0    ; //1mS wait time
(Measure:MWVNA = Disconnect)
            > BasicMeasure  0    ;
repeat 2000     > BasicMeasure  0    ; //200uS wait time
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
            > BasicMeasure  0    ;
```

```
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
        > BasicMeasure  0   ;
halt:
        > BasicMeasure  0   ;
}
```



MWVNAOnly.PAT (vm_MWVNAOnly_A - VM)

TSet	Command	Label	Vector	Measure (MWVNA)	Dig2	Comment
BasicMeasure		start_label MWVNAOnly:	0		0	
BasicMeasure			1	Resync	0	
BasicMeasure	repeat 10		2		0	1uS wait time
BasicMeasure			3	LoadSettingsConnect GSM	0	
BasicMeasure	repeat 50000		4		0	5mS wait time
BasicMeasure			5	Trig GSM	0	
BasicMeasure	repeat 10000		6		0	1mS wait time
BasicMeasure			7	LoadSettingsConnect WCDMA	0	
BasicMeasure	repeat 50000		8		0	5mS wait time
BasicMeasure			9	Trig WCDMA	0	
BasicMeasure	repeat 10000		10		0	1mS wait time
BasicMeasure			11	Disconnect	0	
BasicMeasure	repeat 2000		12		0	200uS wait time
BasicMeasure			13		0	
BasicMeasure			14		0	
BasicMeasure			15		0	

Vertical: 65 Vectors | Vectors 0 to 15

MWVNA Pattern Microcodes Example

< >

UW12GVNA_theory.2.1

<	>

UltraWave 12G VNA Theory Of Operation

UltraWave 12G VNA Theory Of Operation

The UltraWave 12G VNA makes one-port s-parameter measurements using the hardware functions of the UltraWave 12G subsystem.

The following topics are available:

- VNA Basic Operation
- UltraWave 12G VNA Hardware Functions and Signal Flow

For information on MWVNA instrument performance specifications and standards, see UltraWave 12G Specifications.

<	>

UW12GVNA_theory.2.2

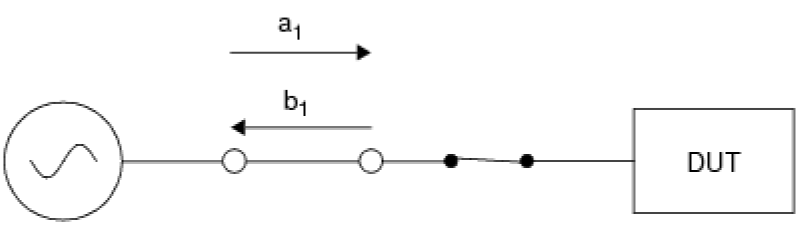
< >

UltraWave 12G VNA Theory Of Operation : VNA Basic Operation

VNA Basic Operation

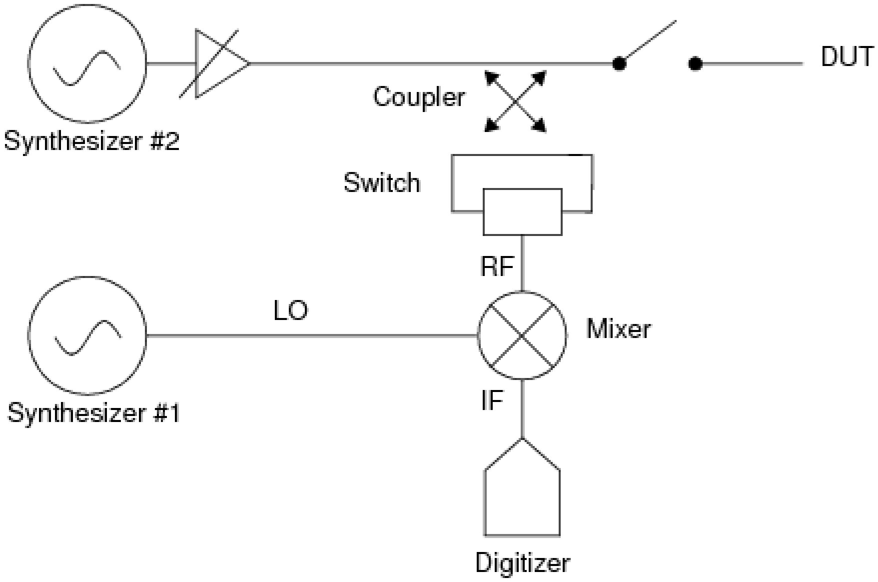
The figure shows a model of a one-port s-parameter measurement. The technique uses a reflection: the forward propagating signal (a_1 - incident to the DUT) and the reverse propagating signal (b_1 - reflected from the DUT) are captured, both as complex numbers. The input reflection coefficient S_{11} is the ratio of the reflected signal b_1 to the forward signal a_1. The value of S_{11} varies with frequency.

$$S_{11} = \frac{b_1}{a_1} \quad (2\text{–}1)$$



One-Port S-Parameter Measurement Model

The following figure shows a basic model of the one-port VNA instrument.



One-Port VNA Instrument Model

To make a one-port s-parameter measurement, the UltraWave 12G VNA performs the following steps:

1.	A signal source generates an incident signal (sine wave).
2.	A directional coupler and switch separate the incident and reflected signals and a mixer downconverts them.
3.	The signals are digitized and processed to yield complex s-parameter values as a function of frequency.
4.	The computed values for S_{11} are stored in a PinListData (PLD) object that contains S1PEx objects (or S1P objects) previously declared and assigned in the test program.

<	>

UW12GVNA_theory.2.3

<	>

UltraWave 12G VNA Theory Of Operation : UltraWave 12G VNA Hardware Functions and Signal Flow

---

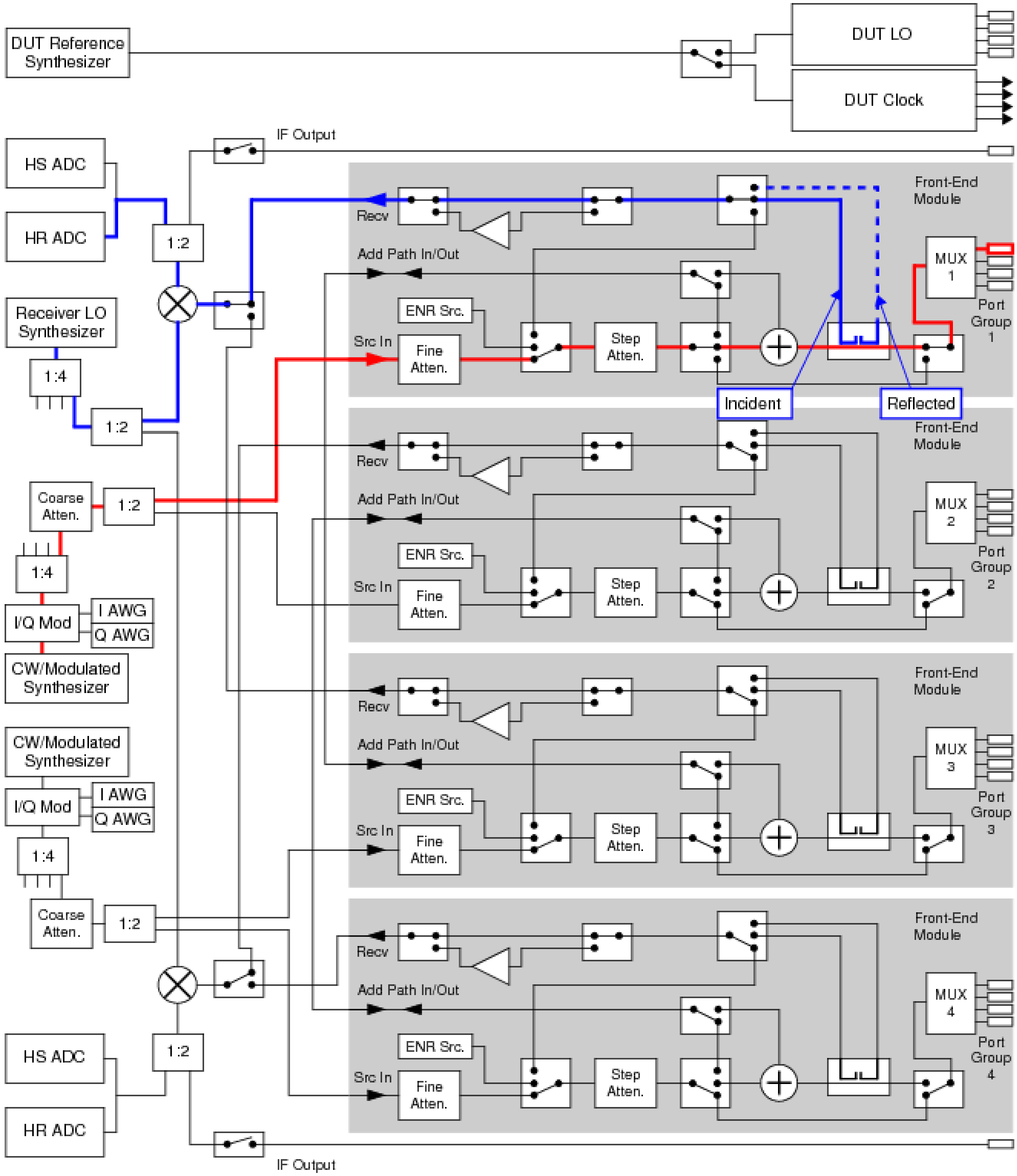
UltraWave 12G VNA Hardware Functions and Signal Flow

The figure shows the configuration of the MW subsystem and received signal flow for the operation and connection of one MWVNA. The second MWReceiver uses the alternate RF input path. Heavy lines represent the signal flow.

Because they share the receiver function with the UltraWave 12G VNA, the UltraWave 12G Receiver cannot make measurements while the UltraWave 12G VNA is operating.

See UltraWave 12G Hardware Functions for more information about the functions in the diagram.

For DIB connections (MW ports, Pogo pins) and signal names for the UltraWave 12G subsystem and instruments, see User View of UltraWave 12G SMPM Connector Blocks, User View of UltraWave 12G DIB Pad Patterns, and UltraWave 12G Signal Descriptions.



MWVNA Hardware Functions and Signal Path

<

>